

SMART ANTI-CORRUPTION AND CITIZEN FEEDBACK PLATFORM USING QR CODE TECHNOLOGY

A Case Study of Rwanda Investigation Bureau (RIB), Kigali, Rwanda

By

LYDIVINE KASINE

2305000371

A research proposal submitted to the University of Kigali in partial fulfillment of the requirements for the award of a Degree with Honors in Bachelor of Business Information Technology, School of Computing and Information Technology.

June 2026

Declaration

I, KASINE Lydivine, declare that this research proposal entitled "Smart Anti-Corruption and Citizen Feedback Platform Using QR Code Technology" is my own original academic work and has been prepared for the award of a Degree with Honors in Bachelor of Business Information Technology at the University of Kigali. This work has not been submitted to any other university or institution for the award of any degree or equivalent qualification.

Signed _____ Date _____

Approval

This is to certify that the research proposal entitled "Smart Anti-Corruption and Citizen Feedback Platform Using QR Code Technology" has been prepared by Kasine Lydivine under my supervision. The proposal has been read, reviewed, and corrected in accordance with academic requirements for research proposal writing at the University of Kigali.

Supervisor: Dr. MUSONI Wilson

Signed _____ Date _____

Table of Contents

Declaration.....	ii
Approval	iii
Table of Contents.....	iv
Abstract.....	vii
List of Figures	viii
List of Tables	ix
List of Abbreviations	x
CHAPTER 1: GENERAL INTRODUCTION AND BACKGROUND OF THE STUDY	11
1.0 Introduction.....	11
1.1 Background of the Study	11
1.1.1 Case Study Context.....	12
1.2 Problem Statement.....	12
1.3 Objectives	13
1.3.1 General Objective	13
1.3.2 Specific Objectives	13
1.4 Research Questions.....	13
1.5 Significance of the Study	13
1.5.1 Significance to Citizens	14
1.5.2 Significance to Public Institutions	14
1.5.3 Significance to Oversight and Leadership	14
1.5.4 Significance to the Researcher and Future Scholars.....	14
1.6 Scope of the Study	14
1.6.1 Time Scope	14
1.6.2 Geographical Scope	14
1.6.3 Content Scope	15
1.7 Limitations of the Study.....	15

CHAPTER 2: LITERATURE REVIEW	16
2.0 Introduction.....	16
2.1 Definition of Key Concepts	16
Citizen Feedback.....	16
Corruption Reporting	16
QR Code Technology	16
Escalation.....	17
Dashboard Monitoring.....	17
Digital Identity Verification.....	17
2.2 Body of the Literature Review.....	17
2.2.1 Empirical Review.....	17
2.2.2 Existing System	20
2.2.3 Proposed System	22
2.3 Theoretical Review	25
2.3.1 Related Theories	25
2.4 Conceptual Framework	25
2.5 Summary of the Gap	27
CHAPTER 3: RESEARCH METHODOLOGY	29
3.0 Introduction.....	29
3.1 Research Design.....	29
3.1.1 Research Approach	30
3.1.2 Research Strategy.....	30
3.1.3 Study Variables.....	31
3.2 Study Area, Population, and Selection of the Sample	31
3.2.1 Sampling Techniques.....	33
3.3 Data Collection Methods	34
Observation.....	35

Interview Guide	35
Documentation Review.....	35
Questionnaire	35
Prototype Review Checklist.....	36
3.4 Data Analysis Methods	36
3.5 Validity and Reliability	37
3.6 Ethical Considerations	37
3.7 Data Collection Procedure	38
3.8 System Requirements Analysis.....	40
3.9 Testing and Prototype Validation Plan	41
References.....	43

Abstract

Corruption, weak complaint follow-up, and unclear service-delivery channels continue to affect the relationship between citizens and public institutions. In many service environments, citizens who experience delay, intimidation, unofficial payment requests, or poor treatment do not always know where to report or how to follow up after reporting. As a result, important complaints are often delayed, ignored, or resolved through informal processes that do not promote accountability.

This proposal presents Chapters One, Two, and Three of a study on a Smart Anti-Corruption and Citizen Feedback Platform Using QR Code Technology with reference to Rwanda Investigation Bureau (RIB), Kigali, Rwanda. The proposed platform is designed to guide citizens to the correct corruption-reporting channel through QR code access, support structured complaint submission, allow evidence attachment and voice-based reporting, strengthen complaint tracking, and improve escalation for unresolved or sensitive cases. Chapter One presents the background, problem statement, objectives, significance, scope, and limitations. Chapter Two reviews key concepts, existing studies, the current reporting system, the proposed system, related theories, and the conceptual framework. Chapter Three explains the methodology that will guide the study, including research design, population and sample, data collection, data analysis, validity, reliability, and ethical considerations.

Keywords: Anti-corruption, citizen feedback, QR code technology, digital governance, public service accountability.

List of Figures

Figure 1: Existing System Flowchart 20

Figure 2: Proposed System Flowchart 23

Figure 3: Conceptual Framework 25

Figure 4: Data Collection and Analysis Procedure 38

List of Tables

Table 1: Comparison between the Existing System and the Proposed System 24

Table 2: Population and Sample Selection 31

Table 3: Data Collection Tools and Purpose 33

Table 4: Functional and Non-Functional Requirements 40

List of Abbreviations

AI: Artificial Intelligence

API: Application Programming Interface

BBIT: Bachelor of Business Information Technology

DPI: Digital Public Infrastructure

ICT: Information and Communication Technology

NIDA: National Identification Agency

QR: Quick Response

RISA: Rwanda Information Society Authority

RIB: Rwanda Investigation Bureau

SACCFP: Smart Anti-Corruption and Citizen Feedback Platform

TAM: Technology Acceptance Model

UI: User Interface

UN: United Nations

UoK: University of Kigali

CHAPTER 1: GENERAL INTRODUCTION AND BACKGROUND OF THE STUDY

1.0 Introduction

This chapter introduces the study and explains the context in which the research problem exists. It presents the background of the study, the problem statement, objectives, research questions, significance of the study, scope, and limitations. The purpose of the chapter is to provide a clear foundation for understanding why a smart anti-corruption and citizen feedback platform is needed in public service institutions and how the proposed study will respond to that need.

Public institutions are expected to provide services in a manner that is transparent, timely, lawful, and respectful to citizens. However, when service delivery is affected by unofficial charging, administrative delay, intimidation, lack of information, or poor complaint follow-up, the consequences go beyond inconvenience. Such weaknesses reduce public trust, discourage citizens from reporting misconduct, and create an environment in which corruption can continue with limited accountability. For this reason, the study considers corruption reporting and citizen feedback not only as administrative activities but also as essential parts of good governance and digital public-service reform.

1.1 Background of the Study

Across the world, governments are increasingly using digital technologies to improve public service delivery, simplify communication with citizens, and strengthen accountability. Digital government platforms, complaint portals, service dashboards, and identity-based systems are now widely recognized as practical tools for reducing administrative friction and improving institutional responsiveness. These developments are particularly important in environments where citizens must move physically between offices to seek services or submit complaints.

In many developing countries, public institutions still face challenges related to weak complaint handling, low transparency, fragmented reporting channels, and inconsistent feedback mechanisms. A citizen may be asked to report verbally, write a complaint on paper, call a phone number, or return on another day without any clear record of the matter submitted. In corruption-sensitive situations, such processes become even weaker because people fear being identified, ignored, or mistreated after reporting. As a result, many incidents remain undocumented, unresolved, or only informally addressed.

Rwanda has made visible progress in digital transformation through e-government services, national identification systems, and wider public access to mobile technology. Even so, corruption-related reporting still requires trusted, accessible, and traceable channels that protect citizens and preserve useful evidence. A person who experiences a bribery request, intimidation, abuse of office, or unfair treatment may fear

reporting through face-to-face channels if confidentiality and follow-up are unclear. In such settings, a weak reporting system creates room for silence, delay, and possible loss of evidence.

1.1.1 Case Study Context

This study uses Rwanda Investigation Bureau (RIB), Kigali, Rwanda, as the case study setting. RIB is an appropriate case because it is a national investigative institution whose mandate is connected to crime reporting, evidence handling, and response to offences, including corruption-related and financial crimes. RIB also emphasizes the use of modern technology and partnerships in preventing, detecting, investigating, and responding to crime threats (RIB, n.d.). Therefore, a QR-enabled complaint and citizen feedback platform fits the institutional need for accessible, traceable, and confidential reporting.

The proposed platform is based on the idea that a citizen should be able to scan an institution-specific QR code connected to RIB reporting channels, access the correct public reporting page immediately, view reporting options, submit a structured corruption-related complaint, attach evidence where necessary, and track the progress of the case. By combining accessibility, complaint logging, escalation, and dashboard oversight into one digital workflow, the study seeks to support more transparent and responsive anti-corruption reporting.

1.2 Problem Statement

Although public institutions provide essential services to citizens, many complaint and feedback processes remain fragmented, manual, and weakly traceable. In several cases, citizens who experience delay, unfair treatment, poor communication, or suspected corruption do not have one clear and trusted channel through which to report the issue and monitor the response. Complaints may be spoken informally, written on paper, sent through uncoordinated contacts, or delayed because the citizen does not know where the complaint should go. This makes follow-up difficult and reduces institutional accountability.

The situation becomes more serious when corruption-related concerns are involved. A citizen may fear that reporting a problem directly at the same office could expose his or her identity or lead to retaliation, delay, or deliberate silence. Where official service procedures and costs are not clearly presented, room is created for unofficial payments, confusion, and abuse. At the same time, institution leaders may lack timely dashboards, escalation tools, or complaint histories that could help them identify unresolved issues and act quickly.

Therefore, there is a need for a digital platform that simplifies access to complaint channels, reduces citizen movement, supports structured and traceable reporting, protects sensitive submissions, and strengthens escalation for unresolved cases. This study proposes a Smart Anti-Corruption and Citizen Feedback Platform using QR code technology as a practical response to that problem.

1.3 Objectives

1.3.1 General Objective

To conduct a case-study-based analysis and design a prototype of a smart anti-corruption and citizen feedback platform that enables citizens to report corruption-related concerns efficiently through QR code access, structured digital reporting, evidence support, and accountable complaint follow-up.

1.3.2 Specific Objectives

- i. To conduct a field-based assessment of the current corruption-reporting and citizen-feedback process related to Rwanda Investigation Bureau (RIB).
- ii. To design a QR code-based access model that directs citizens to the correct institutional reporting point.
- iii. To develop a structured complaint and citizen-feedback workflow that supports text, evidence attachment, and voice-based reporting.
- iv. To implement dashboard monitoring and escalation functions that improve accountability and complaint follow-up.
- v. To demonstrate the proposed platform prototype for citizen reporting, institutional response, and anti-corruption reporting improvement.

1.4 Research Questions

- Q1. What challenges exist in the current corruption-reporting and citizen-feedback process related to Rwanda Investigation Bureau (RIB)?
- Q2. How can a QR code-based access model direct citizens to the correct institutional reporting point?
- Q3. How can a structured complaint and citizen-feedback workflow support text, evidence attachment, and voice-based reporting?
- Q4. How can dashboard monitoring and escalation functions improve accountability and complaint follow-up?
- Q5. How can the proposed platform prototype demonstrate citizen reporting, institutional response, and anti-corruption reporting improvement?

1.5 Significance of the Study

The study is important because it addresses a practical governance problem that affects both citizens and institutions. By proposing a digital platform that combines complaint reporting, service guidance, QR code

access, and accountability tracking, the research contributes to a more transparent and citizen-centered service environment.

1.5.1 Significance to Citizens

Citizens are expected to benefit from an easier and more secure reporting process. Instead of depending on informal directions or repeated office visits, a citizen would be able to access the correct institutional page quickly, submit a complaint in a structured way, and follow the progress of the case. This can reduce fear, save time, and improve confidence in public institutions.

1.5.2 Significance to Public Institutions

Public institutions can benefit from clearer complaint records, improved routing of cases, better monitoring of unresolved matters, and stronger internal accountability. The proposed platform can help staff and managers handle complaints more systematically and provide visible response pathways that reduce confusion and delay.

1.5.3 Significance to Oversight and Leadership

Leaders and oversight bodies may use complaint data, escalation history, and dashboard summaries to identify service-delivery bottlenecks, repeated areas of concern, and possible corruption-risk points. This makes it easier to support evidence-based supervision and more timely administrative action.

1.5.4 Significance to the Researcher and Future Scholars

The study provides the researcher with an opportunity to apply knowledge in business information technology, database design, digital governance, and system design to a real social problem. It also creates a useful academic foundation for future researchers who may wish to extend the work into implementation, evaluation, or broader institutional use.

1.6 Scope of the Study

1.6.1 Time Scope

The study is limited to the 2025/2026 academic period during which the proposal, literature review, data collection planning, and system design activities are undertaken for Chapters One, Two, and Three.

1.6.2 Geographical Scope

The research is focused on Rwanda Investigation Bureau (RIB), Kigali, Rwanda, as the case study environment for analyzing the corruption-reporting problem and proposing the digital solution.

1.6.3 Content Scope

The study concentrates on the design of a smart anti-corruption and citizen feedback platform for the selected RIB case study. It covers institution-specific QR code access, complaint submission, feedback tracking, voice-note support, evidence attachment, dashboard oversight, confidentiality support, and escalation. The study does not extend to nationwide deployment, criminal investigation decisions, or direct integration with all government justice systems at proposal stage.

1.7 Limitations of the Study

One limitation of the study is that it is limited to one case-study institution, Rwanda Investigation Bureau (RIB). Therefore, the findings may not represent all anti-corruption, justice, or public-service institutions in Rwanda, even though they can provide useful lessons for similar citizen-facing reporting environments.

A second limitation is that the work is conducted at proposal and prototype-design stage. The study can demonstrate the proposed workflow and system logic, but it does not include full operational deployment, live integration with RIB systems, or long-term measurement of complaint-resolution and investigation outcomes. Another limitation is that corruption-related research is sensitive. Some respondents may hesitate to share detailed experiences, and some institutional information may not be fully accessible during proposal preparation. The study therefore emphasizes confidentiality, ethical handling of data, and responsible presentation of findings.

CHAPTER 2: LITERATURE REVIEW

2.0 Introduction

This chapter reviews the concepts, studies, systems, and theories related to the proposed Smart Anti-Corruption and Citizen Feedback Platform. The purpose of the review is to understand what has already been done, identify gaps in current complaint and accountability mechanisms, and show how the proposed platform is positioned to respond to those gaps in the context of public service delivery.

2.1 Definition of Key Concepts

Citizen Feedback

Citizen feedback refers to information provided by members of the public about their experience with government or institutional services. In this study, it includes satisfaction, dissatisfaction, delay, denial of service, poor staff conduct, suspected misconduct, and suggestions for improvement. The World Bank (2025) treats digital citizen engagement as part of GovTech because digital channels can help governments involve citizens in service design, service delivery, and public decision-making. Therefore, citizen feedback is not only a complaint; it is also evidence that helps institutions understand whether public services are accessible, fair, responsive, and trusted.

Corruption Reporting

Corruption reporting is the formal communication of bribery requests, abuse of office, unofficial payments, intimidation, or other misconduct to a responsible authority through an accepted channel that can support investigation and follow-up. APEC (2025) explains that anti-corruption technologies should support prevention, detection, investigation, sanction, and recovery processes. For this proposal, corruption reporting therefore means a structured process where a citizen can safely submit a concern and where the institution can preserve the record, verify the issue, respond responsibly, and escalate the case when required.

QR Code Technology

QR code technology uses machine-readable coded images that can be scanned by a mobile phone or compatible device to open a digital destination quickly. De Seta (2023) describes the QR code as an infrastructural gateway because it connects physical places, users, and digital systems. In this study, QR codes are used to guide citizens directly from a physical service point to the correct institutional reporting page without requiring the citizen to search for a long website address or depend on verbal directions from staff.

Escalation

Escalation is the movement of a complaint from one administrative level to another when no timely response is given, when the matter is sensitive, or when the issue requires higher authority for action. In the context of accountability systems, escalation is important because it prevents unresolved complaints from remaining invisible. The OECD (2024a) emphasizes that digital government should improve public-sector responsiveness and reliability; an escalation mechanism supports this goal by creating a route for overdue or serious cases to reach supervisors and oversight officers.

Dashboard Monitoring

Dashboard monitoring refers to the use of visual digital summaries to track complaints, response status, overdue cases, trends, and institutional performance indicators in a way that supports supervision and decision-making. Dashboards are useful because they convert complaint records into management information. They can show how many complaints have been received, how many are pending, how many have been escalated, and which service areas are generating repeated dissatisfaction. This aligns with digital-government literature that highlights data, transparency, and reusable public-sector information as foundations for better decisions (OECD, 2023; UN DESA, 2024).

Digital Identity Verification

Digital identity verification is the process of linking a user to secure registration or identity information in order to improve trust, accountability, and controlled access within a digital system. In Rwanda's digital public infrastructure direction, identity, interoperability, and trusted data exchange are treated as important foundations for connected service delivery (RISA, 2026). For this study, identity verification must be balanced with privacy protection, especially where the complaint concerns corruption, intimidation, or fear of retaliation.

2.2 Body of the Literature Review

2.2.1 Empirical Review

Studies on digital governance consistently show that citizens are more likely to trust public institutions when communication channels are clear, feedback mechanisms are responsive, and complaint processes are traceable. Digital complaint systems, service portals, and public dashboards improve visibility by creating records that can be monitored over time. The UN E-Government Survey 2024 emphasizes that digital transformation should support integrated, inclusive, and sustainable public services rather than isolated online pages (UN DESA, 2024). This is relevant to the present study because the proposed system is not limited to displaying information; it also organizes complaint capture, routing, status monitoring, and escalation.

Research on anti-corruption and whistleblowing platforms demonstrates that the design of the reporting channel strongly affects whether people choose to report. Where channels are difficult to access, poorly explained, or not trusted, citizens are less likely to submit complaints. On the other hand, systems that simplify entry, clarify process steps, and protect the user can encourage participation and improve institutional follow-up. Mattoni and Bratu (2024) show that digital technologies are increasingly discussed as anti-corruption tools, but they also warn that technology must be understood together with institutional actors, public discourse, and measurement practices. This means that a digital platform alone is not enough; it must be connected to clear responsibilities and credible follow-up.

Literature on QR code use in service access shows that QR codes reduce the number of steps required to reach a service page. Rather than typing a long web address or searching through several menus, a citizen can scan a code placed at the institution and open the appropriate page immediately. A usability study by Rivas, Peterson, Schulzetenbers, and Wang (2024) found that respondents were able to use QR codes on smartphones to access a survey quickly and easily. This supports the practical value of QR technology in public service settings where users may have different levels of digital literacy and need quick guidance at the point of service.

The literature also shows the importance of interoperability. Rwanda's recent digital public infrastructure direction emphasizes that standalone platforms can create fragmentation when they cannot communicate with one another, while shared infrastructure can improve integration, inclusion, and service delivery (RISA, 2026). This observation is directly linked to the proposed platform because the system should not operate as an isolated complaint page. It should connect citizen access, institution records, complaint evidence, dashboard monitoring, and escalation into one coordinated process.

Accessibility is another major issue in digital governance. A citizen-facing reporting platform should be usable by people with different levels of literacy, device access, language confidence, and disability status. Voice-supported reporting is therefore important because it can help users who struggle to type long explanations. The W3C Web Accessibility Initiative (2024) explains that speech recognition supports speech-to-text input and other speech user interfaces. In this proposal, voice notes are not treated as decorative features; they are included to make reporting more inclusive and practical.

Despite these developments, a major gap remains in the integration of corruption reporting, institutional access, dashboard monitoring, escalation, confidentiality support, and citizen-friendly communication within one platform. Many existing solutions provide only a portion of the workflow. Some focus on reporting, others on oversight, and others on information display. The present study therefore responds to the need for a more unified and practical platform in the RIB case-study context. The gap is especially important in corruption-reporting environments where citizens need clarity, protection, and follow-up, while institutions need reliable records and management information for accountability.

2.2.2 Existing System

Figure 1: Existing System Flowchart



In corruption-reporting environments related to the RIB case study, the existing complaint process may still depend on physical visits, verbal explanations, phone calls, or fragmented reporting channels. A citizen who experiences or observes a corruption-related problem may first ask where to report, then receive different directions, and finally submit the issue through a channel that may not provide clear tracking. In such a system, the complaint may not receive a reliable case number, status update, or guaranteed follow-up path visible to the reporting citizen.

The weakness of the existing system lies not only in the absence of digitization, but also in the absence of one coherent workflow. Citizens may not know the right office, the correct procedure, or the next step after submission. Leaders may also fail to see which complaints are overdue or unresolved. This creates a gap between complaint submission and actual accountability.

The existing flowchart shows that the current complaint path begins when a citizen faces a service problem and then asks staff members or other citizens where the issue should be reported. The process depends on verbal direction, physical movement between offices, and the availability of the right person at the right time. Because the first contact is not captured in one digital channel, the citizen may repeat the same explanation several times before reaching the correct office.

The figure also shows why follow-up is weak in the manual arrangement. A complaint may be received informally, recorded late, or handled without a visible tracking number. This makes it difficult for citizens to know whether their concern is being reviewed, whether evidence has been kept, or whether the matter has been escalated to a responsible officer. As a result, the existing system can create delays, repeated visits, uncertainty, and limited accountability.

2.2.3 Proposed System

The proposed system is a Smart Anti-Corruption and Citizen Feedback Platform designed to support corruption-reporting transparency and responsive complaint handling. The system uses institution-specific QR codes as entry points so that a citizen can quickly access the correct RIB-related reporting page without searching through multiple channels. From that point, the user can select the appropriate complaint category and submit information in a structured format.

The platform is intended to support citizen registration, complaint submission, evidence attachment, voice-note reporting, case tracking, dashboard visibility, and escalation. This means the system is not only a reporting form but also a complete workflow for handling the complaint from submission to response. The presence of dashboards and escalation logic strengthens management oversight and reduces the risk that complaints will disappear in informal administrative processes.

In practical terms, the proposed system begins at the public institution, where a unique QR code is generated for a specific office or service point. When a citizen scans the code, the platform opens the correct service and reporting page without requiring the citizen to search for a web address or move from one office to another. This direct entry point is important because it reduces confusion at the beginning of the complaint process and makes the reporting channel visible at the place where services are delivered.

The citizen side of the system is designed to make reporting simple but complete. A citizen can provide complaint details, select the related service or category, attach supporting evidence, and use a voice note where written explanation is difficult. The system can also provide a reference or tracking status so that the citizen is not left without feedback after submission. This gives the citizen a clearer relationship with the institution because the report becomes a traceable case instead of an informal conversation.

On the institutional side, the platform supports officers and administrators through organized dashboards. Institution users can view new complaints, read evidence, respond to cases, update the status, and forward unresolved matters for further attention. Administrators can manage users, institutions, complaint categories, and reports. This role separation is important because it allows each user to perform only the tasks connected to their responsibility while still contributing to the same accountability workflow.

The proposed system is therefore more than a digital form. It is a coordinated feedback and anti-corruption support platform that connects QR access, complaint intake, evidence management, response monitoring, escalation, and reporting. The value of the project is that it turns scattered citizen experiences into structured records that can be reviewed by authorized users. This supports transparency because complaints can be followed, measured, and used to identify repeated service problems within the institution.

The proposed design is particularly suitable for the RIB case study because it reduces uncertainty in reporting, improves clarity of access, supports evidence preservation, and creates more reliable institutional records. It also allows the researcher to combine anti-corruption reporting with digital accountability principles in one platform model.

Figure 2: Proposed System Flowchart

The proposed system begins with QR access and ends with traceable complaint handling, institutional response, and escalation where necessary.

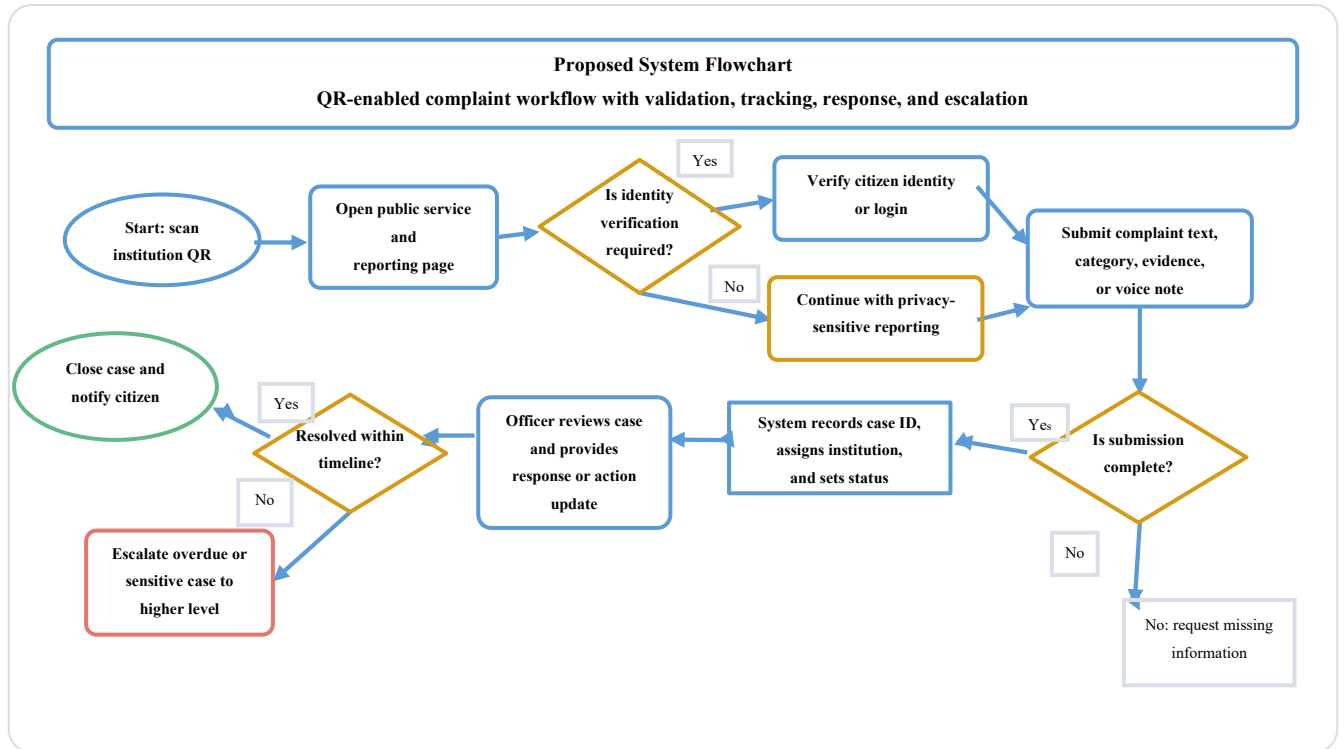


Table 1: Comparison between the Existing System and the Proposed System

The comparison highlights the major operational differences between the current manual complaint approach and the proposed QR-enabled digital platform.

Comparison Area	Existing System	Proposed System
Access to complaint channel	A citizen may depend on asking where to report, calling different contacts, visiting an office physically, or searching for the correct corruption-reporting channel.	A citizen scans a QR code linked to the RIB-related reporting page and reaches the correct complaint channel immediately.
Complaint capture	Complaints may be verbal, phone-based, paper-based, or submitted through fragmented channels, which can weaken evidence preservation and follow-up.	Complaints are submitted through structured digital forms with support for evidence attachment, voice notes, and clear complaint categories.
Tracking and follow-up	Case status may be unclear to the reporting citizen, and follow-up may depend on repeated calls, visits, or informal reminders.	Each complaint is logged with a reference number, tracked through status updates, and made visible to authorized users through monitoring tools.
Escalation	Escalation of unresolved or sensitive corruption-related matters may be delayed if the reporting path is not clearly documented.	Unresolved or sensitive complaints can be escalated through defined digital pathways to authorized supervisors or oversight users.
Institutional oversight	Institutional leaders may have limited real-time	Dashboards support oversight, complaint review,

	visibility of complaint trends, overdue issues, repeated risk areas, and response performance.	trend analysis, and action on overdue or unresolved corruption-reporting issues.
--	--	--

2.3 Theoretical Review

2.3.1 Related Theories

The Technology Acceptance Model (TAM), developed by Davis (1989), is relevant to this study because the success of the proposed system depends on whether citizens and institutional users perceive it as useful and easy to use. If the reporting process is simple, quick, and understandable, adoption is likely to improve. If the system appears difficult, unsafe, or confusing, both citizens and staff may avoid it even when the system is technically sound.

Systems Theory is also important because the proposed platform brings together several interrelated components, including citizens, institutions, QR codes, reporting forms, databases, dashboards, and escalation rules. The value of the platform depends on how well these components work together as one coordinated system. Weakness in one part of the process can affect the whole complaint workflow.

The Information Systems Success Model also supports this study because it links system quality, information quality, service quality, use, user satisfaction, and net benefits (DeLone & McLean, 2003). These dimensions are useful for judging whether the proposed platform is not only technically functional but also meaningful to citizens, officers, and supervisors. For example, a complaint dashboard may exist, but if the information is incomplete or officers do not respond, the system will not produce the accountability benefits expected from it.

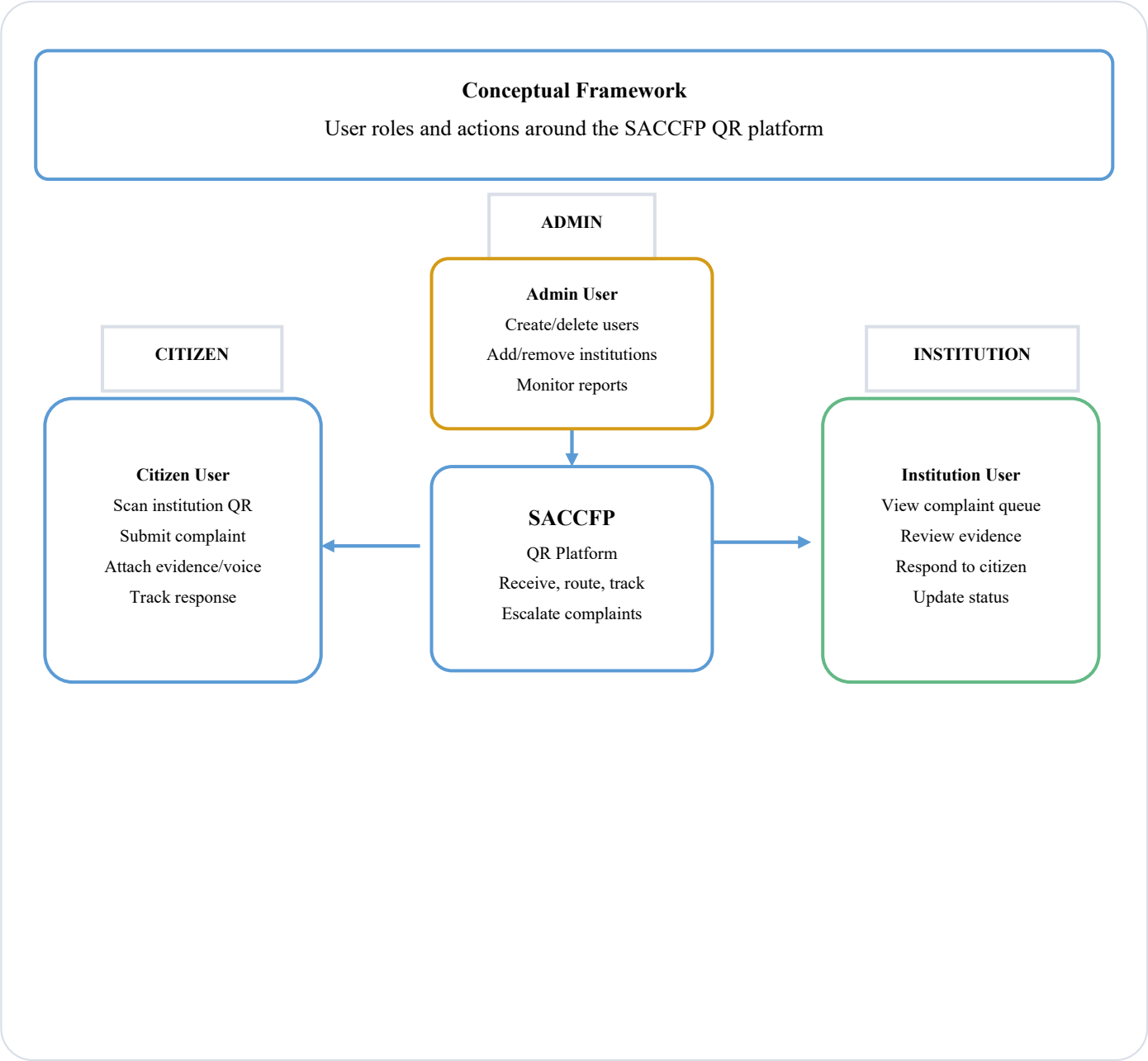
In addition, accountability-oriented thinking supports the study by emphasizing that public institutions should be answerable for service outcomes and should provide clear channels through which complaints can be received, reviewed, and resolved. The proposed platform translates this idea into a practical digital workflow by making complaint handling more visible and measurable.

2.4 Conceptual Framework

The conceptual framework explains the relationship between the main study variables. In this research, QR code access, complaint structure, evidence support, dashboard monitoring, and escalation are treated as key system features that influence transparency, citizen trust, institutional response, and accountability outcomes.

The framework places SACCFP, the QR-enabled Smart Anti-Corruption and Citizen Feedback Platform, at the centre of the model. Citizen, admin, and institution users are arranged around the platform to show how each role interacts with it: citizens report and track complaints, administrators manage users and institutional records, and institution users review, respond, update status, and support escalation.

Figure 3: Conceptual Framework



2.5 Summary of the Gap

The main gap identified in the literature is the absence of one integrated platform that combines easy access, structured complaint intake, evidence support, citizen-friendly design, dashboard monitoring, and escalation. Existing systems frequently address only one or two of these functions at a time, leaving citizens to struggle with fragmented processes and weak follow-up.

The proposed study addresses this gap by presenting a digital platform that connects the citizen, the institution, the complaint record, and the response pathway in one coordinated workflow. In that way, the study contributes both to digital governance practice and to the academic discussion on citizen-centered accountability systems.

CHAPTER 3: RESEARCH METHODOLOGY

3.0 Introduction

This chapter explains the methodology that will guide the study. It describes the research design, research approach, population and sample, the methods for data collection, the method of presenting and analyzing data, and the measures for validity, reliability, and ethical conduct. The chapter is important because it shows how the study will move from the identified problem to a structured and academically justifiable solution.

Methodology is central to this proposal because the study is not only descriptive; it is also design-oriented. The research must first understand the existing complaint and feedback weaknesses, then translate those weaknesses into clear system requirements. For that reason, the chapter combines qualitative and quantitative research methods with system-design analysis. This combination allows the researcher to collect evidence from the service environment while also producing diagrams, workflows, and design specifications that can guide later system development.

3.1 Research Design

The study adopts an applied research design with a system design orientation. It does not only seek to describe the problem of weak complaint follow-up and fragmented reporting in public institutions, but also to propose a practical digital solution. This makes the design appropriate because the study aims at both analysis and problem solving.

The research design combines literature review, observation of the service environment, guided interaction with relevant respondents, and system modeling. Through this approach, the study can identify real challenges in complaint handling while also translating those findings into system requirements, workflows, and platform modules.

The study uses a case-study orientation because the proposed platform is examined in relation to Rwanda Investigation Bureau (RIB). A case study is appropriate when the researcher needs to understand a real reporting environment and connect the findings to a practical intervention. The design is also mixed in nature: qualitative evidence will be used to understand experiences, procedures, fears, and expectations, while quantitative descriptive information will be used to summarize respondent categories, sample distribution, and common reporting patterns. This approach follows the logic of research designs that combine

qualitative interpretation with descriptive measurement when a problem has both human and technical dimensions (Creswell & Creswell, 2023; Saunders, Lewis, & Thornhill, 2019).

3.1.1 Research Approach

The study will use both qualitative and quantitative research methods. The qualitative method will be used to understand experiences, explanations, fears, expectations, and institutional workflow issues related to complaint reporting. It will help the researcher interpret why citizens may avoid reporting, why follow-up becomes weak, and what kind of digital workflow would be trusted by users.

The quantitative method will be used to summarize measurable information from the sample, including respondent categories, frequency of complaint experiences, preferred reporting channels, level of trust in current processes, and willingness to use QR code reporting. Using both methods gives the study a stronger foundation because the qualitative findings explain the meaning of the problem while the quantitative findings show the distribution of the problem among respondents.

This mixed-methods approach is suitable for a final-year IT project because the study must understand people, institutional procedures, and system requirements at the same time. The qualitative part explains the service-delivery problem in depth, while the quantitative part helps the researcher present measurable evidence from the selected sample.

3.1.2 Research Strategy

The research strategy is a case study supported by system design. The case study focuses on Rwanda Investigation Bureau (RIB) as a relevant anti-corruption and investigative reporting environment, while the system-design part translates the findings into a QR-enabled complaint and feedback platform. This strategy is suitable because the study aims to produce a practical IT solution that responds to a real corruption-reporting and citizen-accountability challenge.

The strategy will move from evidence collection to design interpretation. First, the researcher will collect data about current complaint handling and citizen feedback. Second, the findings will be grouped into themes and descriptive summaries. Third, the findings will be converted into system features such as QR access, structured complaint forms, evidence attachment, voice-note support, status tracking, dashboard monitoring, and escalation.

3.1.3 Study Variables

The main input variables of the study are QR code access, citizen complaint details, evidence support, voice-note reporting, identity or privacy route, and institutional service information. The process variable is the digital complaint workflow, which includes validation, routing, response, dashboard monitoring, and escalation. The expected output variables are improved complaint visibility, timely response, reduced confusion, better accountability, and increased citizen trust.

Presenting the variables in this way helps connect the research objectives with the conceptual framework and the proposed system. It also shows that the study is not only describing a problem but linking specific system features to expected public-service outcomes.

3.2 Study Area, Population, and Selection of the Sample

The study will be carried out with reference to Rwanda Investigation Bureau (RIB), Kigali, Rwanda. The area is appropriate because RIB receives and handles crime-related information and is connected to the investigation of corruption-related and financial offences. The case-study area therefore provides a practical setting for examining how a QR-enabled feedback and complaint platform could improve access, traceability, confidentiality, evidence preservation, and accountability in corruption reporting.

The target population for the study includes citizens and public service users who may report corruption-related concerns, selected RIB staff or authorized institutional users, justice and anti-corruption stakeholders, ICT-related reviewers, and community stakeholders who understand citizen reporting. These groups are relevant because each one contributes a different perspective on the reporting process, system usability, confidentiality, evidence handling, and accountability requirements. Citizens explain lived reporting concerns; institutional users explain workflow; oversight stakeholders explain escalation and accountability needs; technical reviewers help judge whether the proposed design is practical.

For proposal planning, the accessible study population is estimated at 400 respondents drawn from people who may interact with RIB reporting channels and selected stakeholders connected to corruption reporting, justice, ICT, and public accountability. This figure is not the total population served by RIB; it is a manageable study population for fieldwork. Since it is not practical to collect data from the whole accessible population, a sample will be

selected. The study uses Yamane's sample-size formula because it is simple and appropriate for a finite population at a known confidence level (Yamane, 1967). The formula is:

$$n = N / (1 + N(e)^2)$$

Where n is the sample size, N is the accessible study population size, and e is the margin of error. Using a margin of error of 5 percent, the sample is calculated as follows:

$$n = 400 / (1 + 400(0.05)^2) = 200$$

The proposed sample size is therefore 200 respondents. The sampling plan combines random sampling and purposive sampling. Random sampling will be used where the researcher needs broad citizen representation, while purposive sampling will be used for selected RIB staff, justice and anti-corruption stakeholders, oversight users, and technical reviewers because those respondents must have specific knowledge of corruption reporting, complaint handling, escalation, confidentiality, or system feasibility.

Table 2: Population and Sample Selection

The table presents the target population categories, proposed sample size, and sampling approach for the study.

Respondent Group	Estimated Accessible Population	Suggested Sample	Sampling Technique
Citizens and public service users who may report corruption-related concerns	300	150	Simple random sampling
Selected RIB staff or authorized institutional users	25	15	Purposive sampling
Justice, anti-corruption, and oversight stakeholders	25	15	Purposive sampling

ICT / security and technical reviewers	10	8	Purposive sampling
Community stakeholders and citizen-reporting observers	40	12	Random and purposive sampling
Total	400	200	Sample size guided by Yamane's formula at 5% margin of error

3.2.1 Sampling Techniques

The study will use random sampling and purposive sampling. Random sampling will be used for citizens and general public service users so that the study can capture a broader range of experiences from people who may need corruption-reporting channels. Purposive sampling will be used for selected RIB staff, justice and anti-corruption stakeholders, oversight users, ICT reviewers, and selected community stakeholders because these groups possess specific knowledge about reporting procedures, complaint handling, escalation, confidentiality, or system feasibility.

Combining the two techniques is appropriate because the study needs both general citizen experience and expert institutional information. The random component improves fairness in selecting ordinary respondents, while the purposive component ensures that the researcher obtains information from respondents who can explain the investigative, accountability, and technical requirements of the proposed platform.

The final sample may be adjusted during fieldwork depending on access, response rate, and approval conditions. However, the planned distribution gives the study a clear balance between ordinary citizen experience and expert institutional knowledge. This balance is important because the proposed platform must be understandable to citizens and workable for the officers who will receive, review, and respond to complaints.

3.3 Data Collection Methods

Data for the study will be collected using both primary and secondary methods. The use of more than one tool is important because the problem under study has both practical and documentary dimensions. Primary methods help the researcher understand the lived experience of reporting and service access, while secondary sources provide scholarly and institutional support for the design of the proposed platform.

The tools are selected to support triangulation. Observation will show what happens in the service environment, interviews will explain why certain problems occur, questionnaires will summarize common user experiences, and documentation review will connect the findings to established research and policy. A prototype review checklist will also be used at design level to check whether the proposed platform responds to the actual complaint-handling needs identified during data collection.

Table 3: Data Collection Tools and Purpose

The table summarizes the tools that will be used to gather research evidence and translate it into system requirements.

Tool	Purpose	Expected Contribution
Observation	To observe how citizens seek services and how complaint handling currently happens.	Helps identify delays, confusion points, and practical workflow weaknesses.
Interview guide	To collect views from citizens and officers on current reporting challenges.	Provides qualitative insights for system requirements and platform features.
Questionnaire	To gather structured responses from citizens and service users on access, trust, delays, and preferred reporting channels.	Supports descriptive analysis using frequencies, percentages, and summarized patterns.

Documentation review	To review books, articles, reports, and official materials related to digital governance and anti-corruption reporting.	Strengthens the literature review and theoretical framework.
Prototype review checklist	To assess whether the proposed interface, workflow, and dashboard design meet identified user and institutional requirements.	Supports validation of the design before full implementation.

Observation

Observation will help the researcher understand how citizens move through service points, how directions are given, and where confusion or delay occurs in the current process. It is useful because some weaknesses are easier to see in practice than to capture through formal documents alone.

Interview Guide

Interviews will provide deeper explanation from respondents about their experiences, expectations, and opinions regarding complaint handling, trust, follow-up, and digital access. The method is appropriate because the proposed platform depends heavily on user needs and institutional realities.

Documentation Review

Documentation review will be used to collect information from journals, reports, books, institutional publications, and other related materials. This tool is important for building the academic foundation of the study and connecting the case study to wider literature on digital governance, QR codes, public accountability, and system development.

Questionnaire

A questionnaire will be used to collect structured information from citizens and service users. It will include questions on ease of access to services, knowledge of complaint channels, trust

in current follow-up, willingness to use QR code access, preferred reporting language, and comfort with evidence or voice-note submission. The questionnaire is useful because it can summarize the opinions of a larger group and reveal patterns that may not appear clearly in interviews alone.

Prototype Review Checklist

The prototype review checklist will be used to assess whether the proposed platform design includes the expected functions: QR access, clear guidance, complaint categories, evidence support, voice-note support, case tracking, dashboard monitoring, and escalation. The checklist will help the researcher connect research findings to the actual system design, which strengthens the practical contribution of the study.

3.4 Data Analysis Methods

The data collected will be presented using descriptive explanations, tables, and thematic summaries. Qualitative analysis will be applied to identify recurring issues in complaint handling, access barriers, accountability gaps, and user expectations. The analysis will then be linked directly to system requirements and platform modules.

This means that if several respondents point to the same problem, such as repeated office visits, lack of response, poor clarity of process, or fear of local exposure, such issues will be interpreted as design requirements for the proposed platform. In this way, the methodology supports a strong connection between research findings and the final digital solution.

Qualitative responses from interviews and observation notes will be analyzed thematically. Thematic analysis is suitable because it allows the researcher to identify repeated patterns across respondent statements and field observations (Braun & Clarke, 2006). Possible themes may include unclear complaint channels, fear of exposure, lack of feedback, repeated physical movement, informal payment pressure, and the need for dashboard visibility. Quantitative questionnaire responses will be presented using frequencies, percentages, and simple tables so that the study can show how common certain experiences or preferences are among respondents.

The final stage of analysis will translate research themes into system requirements. For example, if respondents report that they do not know where to complain, the design response will be QR-based access and service guidance. If respondents report that cases are ignored, the design response will be case tracking and escalation. If officers report that they cannot see

overdue issues, the design response will be dashboard monitoring and alert logic. This method keeps the proposed system grounded in research evidence rather than personal assumption.

3.5 Validity and Reliability

Validity in this study will be supported by ensuring that the tools used for data collection are clearly related to the research objectives and questions. The researcher will also rely on relevant literature and consistent interpretation of the findings so that the proposed design is based on real and meaningful evidence.

Reliability will be strengthened through clear documentation of methods, orderly data handling, and consistent use of concepts throughout the study. The same logic used to define the problem, collect data, and design the system will be maintained across the proposal so that the research process remains coherent and repeatable.

The researcher will also use triangulation by comparing information from observation, interviews, questionnaires, and documents. If different tools point to the same problem, the finding becomes stronger. For example, if citizens mention repeated visits, officers confirm that follow-up is manual, and observation shows unclear service direction, the study can confidently treat weak traceability as a genuine design problem. The instruments will be reviewed before use so that unclear or leading questions can be corrected.

Reliability will further be supported through consistent sampling procedures, careful record keeping, and transparent explanation of how findings are converted into design requirements. This is important because a system-design study should be repeatable: another researcher should be able to understand how evidence led to the chosen modules, diagrams, and workflow decisions.

3.6 Ethical Considerations

Ethical care is especially important in a study related to corruption and public complaint reporting because the topic may involve sensitive experiences, vulnerable respondents, and trust concerns. The study will therefore observe strong ethical principles during data collection and presentation.

- Respondents will be treated with respect and their participation will be voluntary.
- Personal identities and sensitive complaint details will be handled confidentially.

- Information collected will be used for academic purposes only.
- No allegation will be presented as a confirmed fact without proper evidence and authorization.
- The proposed platform design will aim to reduce retaliation risk against reporting citizens.

By observing these principles, the study will protect respondents, maintain academic integrity, and support responsible discussion of a sensitive but important governance issue.

The study will avoid naming individuals or presenting unverified allegations as facts. Where examples are needed, they will be described generally and used only to explain system requirements. This approach protects participants and keeps the academic focus on improving complaint mechanisms, not accusing specific persons without due process.

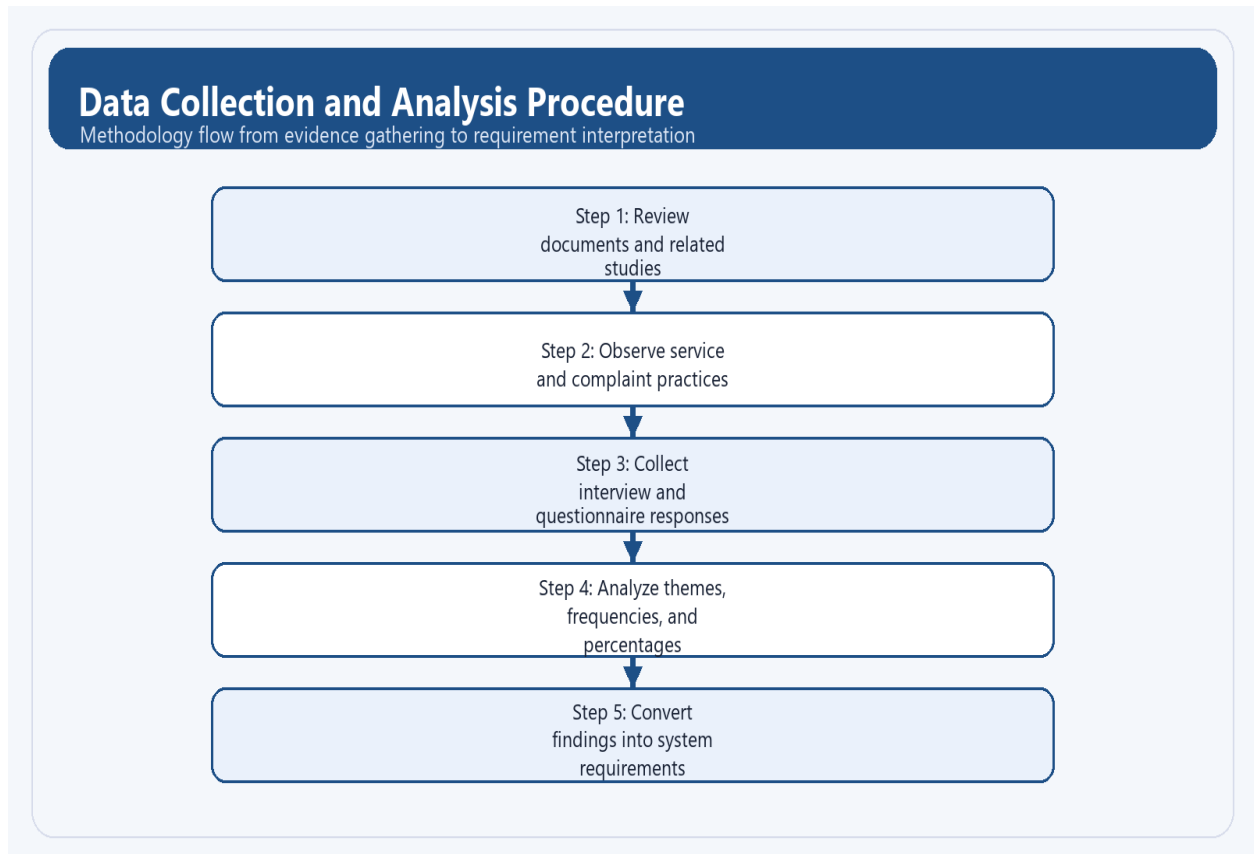
3.7 Data Collection Procedure

The data collection procedure will begin with preparation of the instruments. The researcher will prepare the observation checklist, interview guide, questionnaire, and prototype review checklist. These tools will be aligned with the research objectives so that each question or observation item contributes directly to the problem statement, research questions, or system design requirements. Approval and permission will be sought before engaging respondents.

The first field activity will be observation of the service environment. The researcher will observe how citizens move within the service area, how they ask for directions, where delays appear, and whether complaint guidance is visible. The second activity will involve interviews with selected respondents who understand the current complaint process. The third activity will be questionnaire administration among citizens and service users. Finally, documents and official materials will be reviewed to support the academic and institutional explanation of the proposed platform.

Figure 4: Data Collection and Analysis Procedure

The procedure shows how evidence will move from documents, observation, and respondent input into requirement analysis and design interpretation.



After collection, data will be organized according to source and theme. Observation notes will be grouped around service access, complaint movement, waiting points, and visible guidance. Interview responses will be grouped according to recurring issues such as trust, privacy, escalation, and institutional response. Questionnaire responses will be summarized using descriptive statistics. The combined findings will then be used to refine the system workflow, database entities, user roles, and interface priorities.

3.8 System Requirements Analysis

System requirements analysis is the process of translating research findings into clear statements of what the proposed platform must do. In this study, requirements will be derived from three sources: the literature review, the current complaint process, and the responses from selected users and institutional stakeholders. The requirements will be divided into functional and non-functional requirements.

Functional requirements describe the actions the system must support. The proposed platform must allow citizens to scan an institution-specific QR code, open the correct public reporting page, create or verify a user profile where required, submit a corruption-related complaint, attach evidence, record or upload a voice note, receive a case number, track case status, and receive response updates where appropriate. Authorized officers must be able to view assigned cases, review evidence, respond to complaints, update status, request additional information, and escalate unresolved or sensitive cases. Administrators must be able to manage institutions, authorized users, reporting categories, QR codes, dashboards, and reporting summaries.

Non-functional requirements describe the quality conditions the system must satisfy. The platform should be usable, secure, reliable, responsive on mobile devices, accessible to users with different literacy levels, and capable of protecting sensitive complaint information. It should use clear language, minimize unnecessary steps, and provide visible confirmation after submission. These requirements are important because anti-corruption reporting depends on trust; if the system feels unsafe, confusing, or unreliable, citizens may avoid it.

Table 4: Functional and Non-Functional Requirements

The table connects the proposed platform requirements to the complaint and accountability problems identified in the study.

Requirement Category	Specific Requirement	Purpose in the Study
Access	Institution-specific QR code and public service page	Reduces confusion and guides citizens to the correct reporting point.
Complaint intake	Structured form with categories, description, evidence, and voice-note support	Improves completeness of complaints and preserves useful supporting information.
Tracking	Case number, visible status, and response history	Improves transparency and reduces repeated physical follow-up.
Escalation	Overdue and sensitive-case escalation rules	Prevents unresolved complaints from remaining hidden at one administrative level.
Security and privacy	Role-based access, controlled data visibility, and confidential handling of sensitive cases	Protects citizens and supports responsible institutional use of complaint data.

3.9 Testing and Prototype Validation Plan

The proposal includes a validation plan to show how the prototype will be checked before full implementation. Testing will focus on usability, functional completeness, security logic, and data accuracy. Usability testing will examine whether users can scan a QR code, understand the reporting page, submit a complaint, and track status without unnecessary confusion. Functional testing will check whether each expected feature performs its role: registration, login, complaint submission, evidence attachment, voice-note submission, officer response, dashboard visibility, and escalation.

Security and privacy validation will examine whether role-based access protects sensitive information. Citizens should not see internal officer notes, officers should only see cases assigned or authorized for their role, and administrators should access oversight information responsibly. Data validation will check whether the complaint record keeps the correct category, date, institution, status, evidence link, and response history. These checks are important because weak data quality would undermine accountability.

Prototype validation will also involve feedback from expected users. Citizens can comment on whether the platform is easy to understand and whether they would trust it for reporting. Officers can comment on whether the dashboard supports their work. Supervisors can comment on whether escalation and reporting summaries are useful for oversight. The researcher will use this feedback to refine the final design direction.

References

The reference list combines current digital-governance sources with established research methodology sources used to support the proposal's concepts, methods, design model, and system requirements.

1. APEC Policy Support Unit. (2025). Technologies for Preventing, Detecting, and Combating Corruption. Retrieved from <https://www.apec.org/publications/2025/06/technologies-for-preventing--detecting--and-combating-anti-corruption>
2. Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. *Qualitative Research in Psychology*, 3 (2), 77-101. Retrieved from <https://doi.org/10.1191/1478088706qp063oa>
3. Creswell, J. W., & Creswell, J. D. (2023). *Research design: Qualitative, quantitative, and mixed methods approaches* (6th ed.). SAGE.
4. Davis, F. D. (1989). Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS Quarterly*, 13 (3), 319-340. Retrieved from <https://doi.org/10.2307/249008>
5. de Seta, G. (2023). QR code: The global making of an infrastructural gateway. *Big Data & Society*. Retrieved from <https://journals.sagepub.com/doi/10.1177/20594364231183618>
6. DeLone, W. H., & McLean, E. R. (2003). The DeLone and McLean model of information systems success: A ten-year update. *Journal of Management Information Systems*, 19 (4), 9-30. Retrieved from <https://doi.org/10.1080/07421222.2003.11045748>
7. Mattoni, A., & Bratu, R. (2024). Digital technologies and anti-corruption: Reflections on public discourses, actors' interactions, and the measurement of corruption. *American Behavioral Scientist*. Retrieved from <https://journals.sagepub.com/doi/10.1177/00027642241268531>
8. OECD. (2023). 2023 OECD Open, Useful and Re-usable Data (OURdata) Index: Results and key findings. Retrieved from https://www.oecd.org/en/publications/2023-oecd-open-useful-and-re-usable-data-ourdata-index_a37f51c3-en.html

9. OECD. (2024a). 2023 OECD Digital Government Index: Results and key findings.
Retrieved from <https://www.oecd.org/governance/2023-oecd-digital-government-index-1a89ed5e-en.htm>
10. OECD. (2024b). Anti-Corruption and Integrity Outlook 2024. Retrieved from
https://www.oecd.org/content/dam/oecd/en/publications/reports/2024/03/anti-corruption-and-integrity-outlook-2024_6e7ad8ce/968587cd-en.pdf
11. OECD. (2024). Global Trends in Government Innovation 2024. Retrieved from
https://www.oecd.org/en/publications/global-trends-in-government-innovation-2024_c1bc19c3-en/full-report.html
12. RISA. (2025). Rwanda Launches the Center for Digital Public Infrastructure: A New Era of Innovation and Inclusion. Retrieved from <https://www.risa.gov.rw/news-detail/rwanda-launches-the-center-for-digital-public-infrastructure-a-new-era-of-innovation-and-inclusion>
13. RISA. (2026). Rwanda advances digital future with launch of national DPI strategy.
Retrieved from <https://www.risa.gov.rw/news-detail/rwanda-advances-digital-future-with-launch-of-national-dpi-strategy>
14. Rivas, M. A., Peterson, A., Schulzetenberg, A., & Wang, Y. (2024). Usability of Quick Response (QR) codes as a method to access a survey using a smartphone. U.S. Census Bureau. Retrieved from <https://www.census.gov/library/working-papers/2024/adrm/rsm2024-05.html>
15. Saunders, M., Lewis, P., & Thornhill, A. (2019). Research methods for business students (8th ed.). Pearson.
16. United Nations Department of Economic and Social Affairs. (2024). UN E-Government Survey 2024. Retrieved from <https://desapublications.un.org/publications/un-e-government-survey-2024>
17. W3C Web Accessibility Initiative. (2024). Speech Recognition. Retrieved from <https://www.w3.org/WAI/perspective-videos/voice/>
18. World Bank. (2025). Consultation on Civic and Citizen Engagement Framework and Scorecard Indicator. Retrieved from <https://www.worldbank.org/en/topic/citizen-engagement/brief/consultation-on-civic-and-citizen-engagement-framework-and-scorecard-indicator>

19. World Bank. (2025). GTMI 2025 Update. Retrieved from
<https://www.worldbank.org/en/programs/govtech/gtmi-2025-update>
20. Rwanda Investigation Bureau. (n.d.). What we investigate. Retrieved from
<https://www.rib.gov.rw/what-we-investigate-1>
21. Yamane, T. (1967). Statistics: An introductory analysis (2nd ed.). Harper & Row.